Scientific Computation Project 2

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Part 1

1.

(a-1) This is a nonlinear system of equations. For the non-trivial equilibrium, we need to find solutions numerically using the root function in scipy.optimize with the default (hybr) method. The system of equations are specified using the provided model code after removing the t parameter. A "guess" for the solution is also given. Then the iterative approach moves from the guess towards a solution one step at a time.

(a-2) Once the equilibrium is found, we linearize the system about this equilibrium to get the Jacobian matrix J. We then compute the matrix exponential e^{JT}, and find the initial perturbation vector $\tilde{x}_0$ that leads to the largest energy growth ratio $\frac{e(T)}{e(0)}$. This is done by computing the singular value decomposition (SVD) of e^{JT} and picking the singular vector associated with the largest singular value.

(b) We test consistency by taking the initial perturbation vector $\tilde{x}_0$ found in (a), simulating the full nonlinear system from $x = \bar{x} + \tilde{x}_0$, and then comparing the observed ratio $e(T)/e(0)$ from the nonlinear simulation with the linearized prediction. The outputs, "Linear prediction of max e(T)/e(0) in part1q1a: 26879.887079694287, Nonlinear simulation e(T)/e(0) in simulate1: 26870.19029856483", shows the consistency between these two methods.

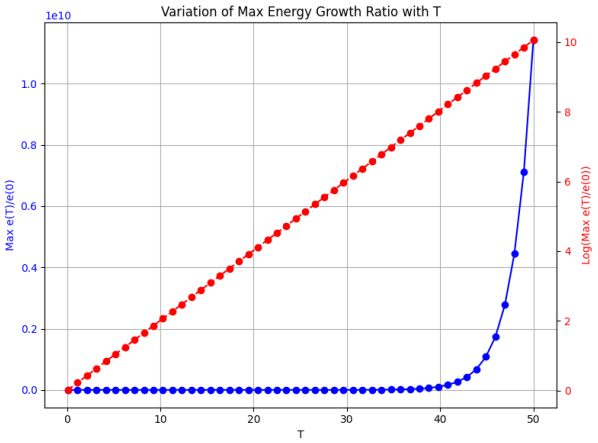
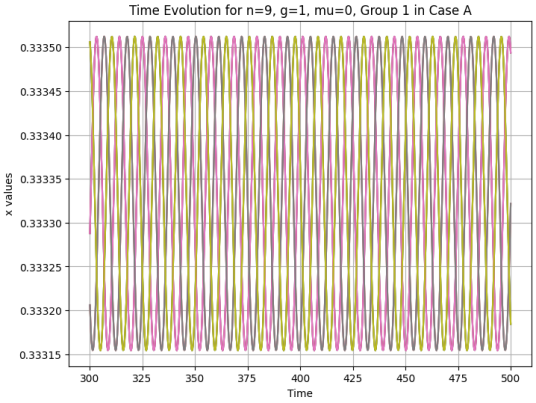


Figure 1: Exponential growth with increasing timestep

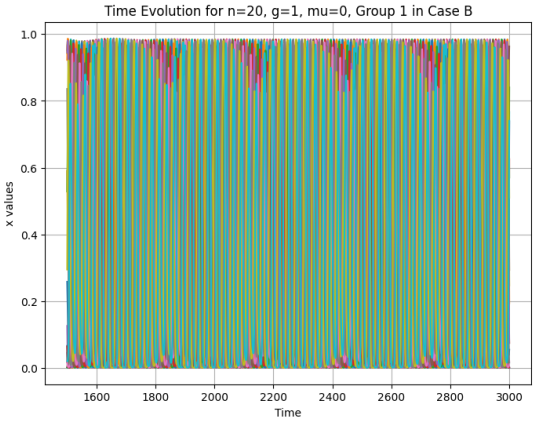
(c) The system exhibits exponential instability over time, with perturbations growing significantly as T increases. $\gamma = 2$ amplifies coupling effects between system components and promotes instability via increasing the magnitude of unstable eigenvalues in J. With $\mu = 0$, stabilizing terms in the system are removed, making it easier for instability to dominate.

2.

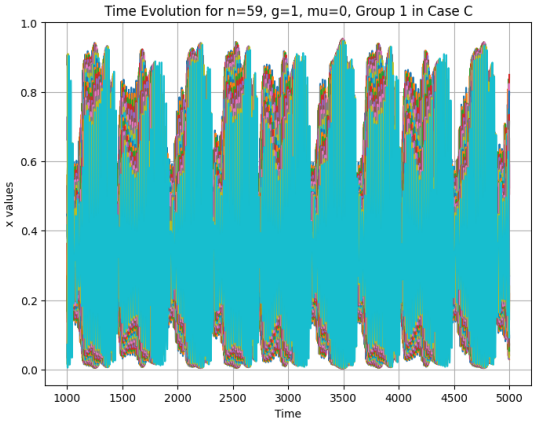
(case A: n = 9) The long-term dynamics are periodic and well-ordered, with no evidence of chaos. The system appears to settle into stable sinusoidal oscillations. The global qualitative dynamics are analysed below:



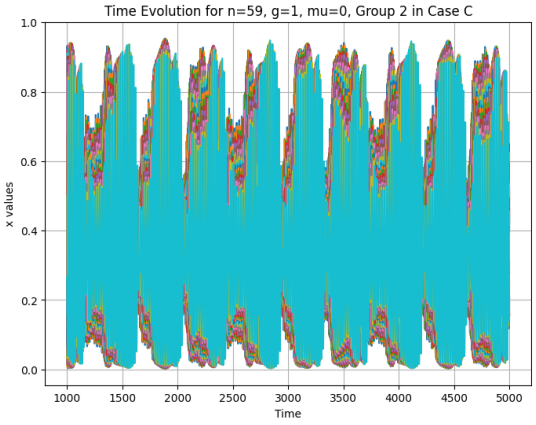
(a) case A: $n = 9$



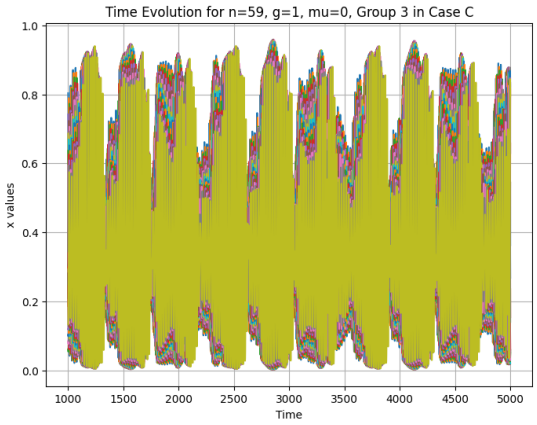
(b) case B: $n = 20$



(c) Group 1 with first 20 components



(d) Group 2 with second 20 components



(e) Group 3 with last 19 components

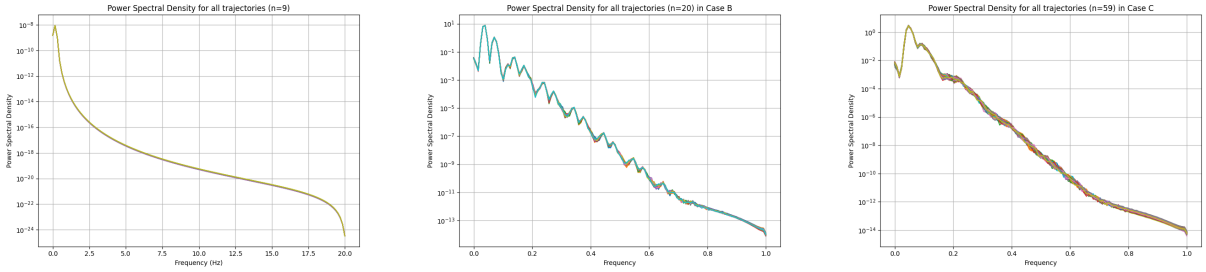
Figure 2: Simulation results for three cases with $\gamma = 1$ and $\mu = 0$

- Time-dependence: Yes. The trajectories clearly exhibit time-dependent behavior, with oscillations occurring over time, shown in Figure 2a.
- Simple sinusoidal oscillations: Yes, the trajectories in Figure 2a exhibit simple sinusoidal oscillations. Each component x_i appears to follow a regular and periodic sinusoidal pattern. The system exhibits strong low-frequency behavior, with diminishing contributions from higher frequencies, shown in Figure 4a.
- Synchronized components: Yes, the components are synchronized. All trajectories maintain a coherent phase and amplitude relationship, indicating strong coupling between the components, supported by the overlapped line in Figure 3a.

- Chaotic dynamics: No chaotic dynamics are present. The regular and sharply decaying PSD is characteristic of periodic motion, shown in Figure 3a.

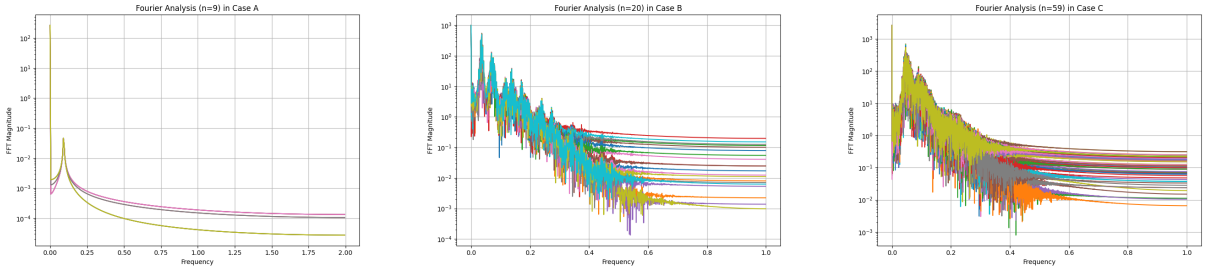
(case B: n = 20) The global qualitative dynamics are analysed below:

- Time-dependence: Yes, the system is time-dependent, as evident from the oscillatory trajectories in Figure 2b.
- Simple sinusoidal oscillations: the trajectories show sinusoidal oscillations, but the oscillations are less uniform than in Case A (n=9). There is noticeable variation in amplitude and phase across the components.
- Synchronized components: Partial synchronization is observed. While the components maintain similar frequencies, there are phase shifts and amplitude differences among them, reducing the overall coherence compared to Case A (n=9).
- Chaotic dynamics: Yes. Figure 4b shows that energy is distributed across a wide range of frequencies rather than being concentrated at a few discrete peaks (as would occur in periodic systems). This broad spectral distribution is a hallmark of chaotic systems.



(a) case A: $n = 9$ (b) case B: $n = 20$ (c) case C: $n = 59$

Figure 3: Power Spectral Density (PSD) results for three cases with $\gamma = 1$ and $\mu = 0$



(a) case A: $n = 9$ (b) case B: $n = 20$ (c) case C: $n = 59$

Figure 4: Fourier analysis results for three cases with $\gamma = 1$ and $\mu = 0$

(case C: n = 59) The oscillation amplitudes increase compared to Cases A and B, reflecting the greater degrees of freedom and interactions in the larger system.

- Time-dependence: Yes, oscillatory trajectories shows the system is time-dependent.
- Simple sinusoidal oscillations: No, sinusoidal oscillations are observed, but with more variability in amplitude and phase compared to Cases A and B. The oscillations are not uniform, and some trajectories exhibit larger deviations.
- Synchronized components: No. The trajectories show noticeable phase differences and amplitude variations, indicating a weakening of global coherence as n increases. There are variations in the magnitude of high-frequency components, implying a lack of perfect synchronization, shown in Figure 3c.

- Chaotic dynamics: Yes, evidence by fourier analysis in Figure 4c. The absence of distinct periodic components (sharp peaks) and the complexity of the dynamics (broad spectrum) are strong indicators of chaos.

Part 2

1. Efficiency

Efficient reasons in matrix memory and solver: The matrix in the dualfd1 function is dense with the shape $2n \times 2n$, while the matrix in the part2q1 function is sparse with the shape $7 \times 2n$, reducing excessive memory due to storing the zeros in A and reducing excessive operations due to the zeros. During our working process, we tried the spslove version which solves the memory issue, but finally we chose the solve_banded which is specially designed for banded matrices with actually using a Fortan routine from the Lapack library. During our process, we use the inefficient dualfd1 to check the correctness of our method due to the solve_banded requires a tricky matrix input.

2. Comparison between implicit part2q1 and 2nd finite difference

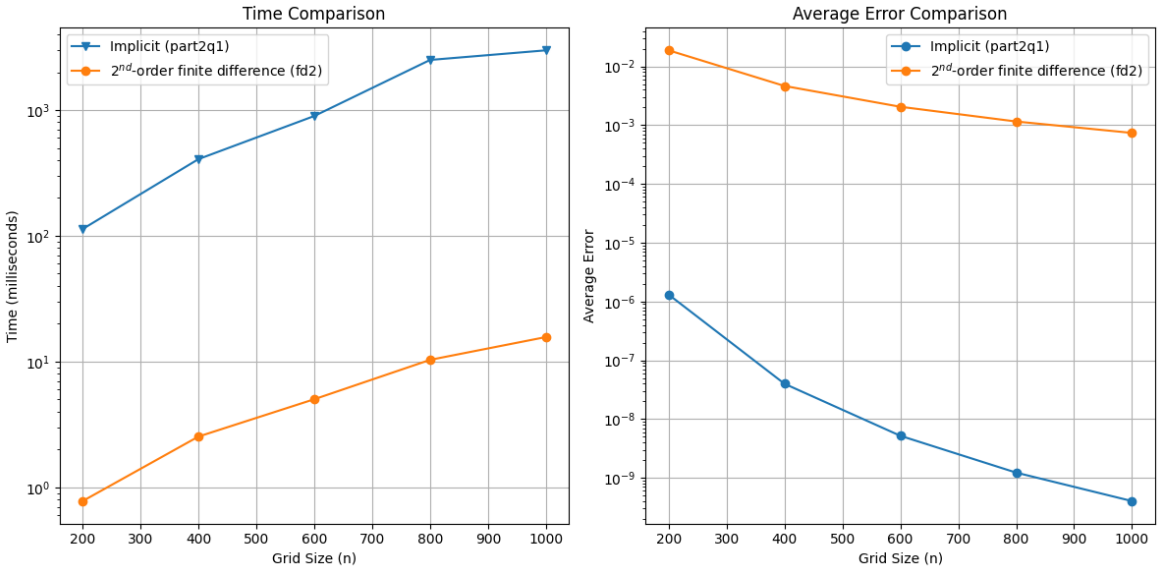


Figure 5: Running time (log) and errors of implicit (part2q1) and 2nd finite difference (fd2)

The test function is $f(x, y) = \sin(k_1 x) + 0.5k_1 \sin(2k_1 x)$ and $k_1 = 8 * \pi/5$ represent the frequencies. The Grid size ranges from 200 to 1000 with a step in 200. Because m is comparable to n, so we keep m = n for simplicity.

Time. Implicit schemes are computationally expensive (as seen in the time comparison), because they require solving a system of equations; the 2nd finite difference scheme is computationally cheaper. The running time of implicit part2q1 increases exponentially, shown in linear trend in the log timescale in Figure 5, while the finite difference method remains efficient even for large grid sizes, n.

Accuracy. Initially, the 2nd finite difference scheme shows higher errors, which decreases as the grid size increases, whereas thes implicit method starts with lower errors. The implicit method's better handling of global features and higher-order terms provides superior accuracy in these areas, supported by faster decreasing ratio from 10^{-6} to 10^{-9}. If the error on the scale of 10^{-3} is acceptable, the finite difference scheme can be used. Otherwise, the implicit method should be chosen, albeit with a higher computational cost.